

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Experiment No.: 5 Name: Vansh Mahant PRN: 24070521038 Section: A Batch: A1

1. Experiment Title:

Array Min/Max Finder & Shopping Cart Calculator — a tabbed web page using HTML, CSS and JavaScript. The first tab finds the minimum and maximum of a dynamically built array using a for-loop; the second tab implements a shopping cart that uses array methods (push, forEach, reduce, filter) to calculate totals, apply tiered discounts, and list expensive products.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, JavaScript

3. Experiment Program Code:

[index.html](#)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Practical 6: Array Finder & Shopping Cart</title>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <div class="app-wrapper">

    <div class="tab-navigation">
      <button id="tab-finder" class="tab-btn active">12 Array Min/Max Finder</button>
      <button id="tab-cart" class="tab-btn">🛒 Shopping Cart Calculator</button>
    </div>

    <div id="finder-container" class="tab-content active">
      <div class="finder-card">
        <h1 class="main-title">Array Min & Max Finder</h1>

        <div class="card controls-card">
          <input type="number" id="number-input" placeholder="Enter a number">
          <div class="button-group">
            <button id="add-number-btn" class="btn btn-blue">Add Number</button>
            <button id="add-random-btn" class="btn btn-teal">Add Random</button>
            <button id="reset-btn" class="btn btn-red">Reset</button>
          </div>
        </div>

        <div class="card array-card">
          <h2>Current Array</h2>
          <div id="array-pills-container" class="pills-container">
            <!-- Badges render here dynamically -->
          </div>
        </div>

        <div class="stats-grid">
          <div class="card stat-card">
            <h3>MAXIMUM VALUE</h3>
          </div>
        </div>
      </div>
    </div>
  </div>
</body>
</html>
```

```

        <p id="max-val-display">-</p>
    </div>
    <div class="card stat-card">
        <h3>MINIMUM VALUE</h3>
        <p id="min-val-display">-</p>
    </div>
    <div class="card stat-card">
        <h3>TOTAL ELEMENTS</h3>
        <p id="total-elements-display">0</p>
    </div>
</div>
</div>

<div id="cart-container" class="tab-content">
    <div class="cart-card-wrapper">
        <h1 class="main-title">Shopping Calculator</h1>

        <!-- Input Form Section -->
        <div class="input-form">
            <input type="text" id="product-name" placeholder="Product Name">
            <input type="number" id="product-price" placeholder="Price" step="0.01">
            <input type="number" id="product-qty" placeholder="Quantity">
            <button id="add-product-btn">Add Product</button>
        </div>

        <!-- Table Section -->
        <div class="table-responsive">
            <table id="cart-table">
                <thead>
                    <tr>
                        <th>ID</th>
                        <th>Product</th>
                        <th>Price</th>
                        <th>Quantity</th>
                        <th>Total</th>
                    </tr>
                </thead>
                <tbody>
                    <!-- Dynamic rows here -->
                </tbody>
            </table>
        </div>

        <!-- Bill Results Card -->
        <div class="bill-summary-card">
            <div class="total-row">
                Total Amount : <span id="total-amount">₹0.00</span>
            </div>
            <div class="discount-row">
                Discount (<span id="discount-percent">0%</span>) : <span id="discount-amount">-
₹0.00</span>
            </div>
            <div class="final-row">
                Final Amount : <span id="final-amount">₹0.00</span>
            </div>
        </div>

        <!-- Lists Display Grid -->
        <div class="lists-container">
            <div class="summary-section">
                <h2>Item Summary</h2>
                <ul id="item-summary-list">
                    <!-- Summary items here -->
                </ul>
            </div>

            <div class="summary-section">

```

```

        <h2>Expensive Products (Price > 1000)</h2>
        <ul id="expensive-products-list">
          <!-- Expensive items here -->
        </ul>
      </div>
    </div>
  </div>
</div>

<script src="script.js"></script>
</body>
</html>

```

styles.css

```

/* Basic Global Styles */
body {
  font-family: Arial, sans-serif;
  background-color: #f8fafc;
  color: #333;
  margin: 0;
  padding: 20px;
  display: flex;
  justify-content: center;
}

.app-wrapper {
  width: 100%;
  max-width: 800px;
  box-sizing: border-box;
}

/* Tab Navigation Styling */
.tab-navigation {
  display: flex;
  margin-bottom: 15px;
  border-bottom: 2px solid #cbd5e1;
}

.tab-btn {
  flex: 1;
  padding: 12px;
  background-color: #e2e8f0;
  border: none;
  border-top-left-radius: 6px;
  border-top-right-radius: 6px;
  font-size: 15px;
  font-weight: bold;
  color: #475569;
  cursor: pointer;
  transition: all 0.2s ease-in-out;
  border: 1px solid #cbd5e1;
  border-bottom: none;
  margin-right: 4px;
}

.tab-btn:hover {
  background-color: #cbd5e1;
}

.tab-btn.active {
  background-color: #ffffff;
  color: #00008b;
  border-bottom: 2px solid #ffffff;
  margin-bottom: -2px;
  box-shadow: 0 -2px 5px rgba(0,0,0,0.05);
}

```

```

/* Tab Content Toggle */
.tab-content {
  display: none;
}

.tab-content.active {
  display: block;
}

/* Common Container Card Styles */
.finder-card, .cart-card-wrapper {
  background-color: #ffffff;
  border: 1px solid #cbd5e1;
  border-radius: 6px;
  padding: 20px;
  box-sizing: border-box;
}

/* Title Header */
.main-title {
  text-align: center;
  color: #00008b;
  font-size: 26px;
  margin-top: 0;
  margin-bottom: 20px;
}

.card {
  background-color: #ffffff;
  border: 1px solid #cbd5e1;
  border-radius: 6px;
  padding: 15px;
  margin-bottom: 20px;
  box-sizing: border-box;
}

/* =====
1. Array Min & Max Finder Styles
===== */

/* Controls Card */
.controls-card {
  display: flex;
  gap: 15px;
  align-items: center;
  flex-wrap: wrap;
}

.controls-card input {
  flex: 1;
  min-width: 150px;
  padding: 8px 12px;
  font-size: 14px;
  border: 1px solid #cccccc;
  border-radius: 4px;
  outline: none;
}

.button-group {
  display: flex;
  gap: 8px;
}

/* Custom Buttons styling */
.btn {
  border: none;
  border-radius: 4px;
  padding: 8px 16px;

```

```

    font-size: 14px;
    font-weight: bold;
    cursor: pointer;
    color: #ffffff;
    transition: background-color 0.2s;
}

.btn-blue {
    background-color: #3b66d4;
}

.btn-blue:hover {
    background-color: #2b51b3;
}

.btn-teal {
    background-color: #0d9488;
}

.btn-teal:hover {
    background-color: #0f766e;
}

.btn-red {
    background-color: #dc2626;
}

.btn-red:hover {
    background-color: #b91c1c;
}

/* Array Badge Container */
.array-card h2 {
    font-size: 16px;
    color: #00008b;
    margin-top: 0;
    margin-bottom: 10px;
    border-bottom: 1px solid #cbd5e1;
    padding-bottom: 6px;
}

.pills-container {
    border: 1px solid #cbd5e1;
    background-color: #ffffff;
    border-radius: 4px;
    padding: 15px;
    min-height: 40px;
    display: flex;
    flex-wrap: wrap;
    align-items: center;
    gap: 8px;
}

/* Badge/Pill Style */
.pill {
    min-width: 32px;
    height: 32px;
    padding: 0 8px;
    border: 1px solid #cbd5e1;
    border-radius: 500px;
    background-color: #ffffff;
    font-weight: bold;
    display: flex;
    justify-content: center;
    align-items: center;
    font-size: 14px;
    box-shadow: 0 1px 2px rgba(0, 0, 0, 0.05);
}

```

```

/* 3-Column Stats Grid */
.stats-grid {
  display: grid;
  grid-template-columns: 1fr 1fr 1fr;
  gap: 15px;
  margin-bottom: 20px;
}

.stat-card {
  text-align: center;
  background-color: #f8fafc;
  padding: 12px;
  margin-bottom: 0;
}

.stat-card h3 {
  font-size: 11px;
  font-weight: bold;
  color: #64748b;
  margin-top: 0;
  margin-bottom: 6px;
  letter-spacing: 0.5px;
}

.stat-card p {
  font-size: 24px;
  font-weight: bold;
  color: #00008b;
  margin: 0;
}

/* =====
2. Shopping Cart Calculator Styles
===== */

/* Horizontal Inputs & Add Button */
.input-form {
  display: flex;
  gap: 10px;
  margin-bottom: 20px;
}

.input-form input {
  flex: 1;
  padding: 8px 12px;
  font-size: 14px;
  border: 1px solid #cccccc;
  border-radius: 4px;
  outline: none;
}

.input-form button {
  background-color: #3b66d4;
  color: #ffffff;
  border: none;
  border-radius: 4px;
  padding: 8px 16px;
  font-size: 14px;
  font-weight: bold;
  cursor: pointer;
  transition: background-color 0.2s;
}

.input-form button:hover {
  background-color: #2b51b3;
}

/* Bordered Data Table */

```

```

.table-responsive {
  overflow-x: auto;
}

#cart-table {
  width: 100%;
  border-collapse: collapse;
  margin-bottom: 20px;
}

#cart-table th, #cart-table td {
  padding: 10px;
  text-align: center;
  border: 1px solid #cbd5e1;
  font-size: 14px;
}

#cart-table th {
  background-color: #3b66d4;
  color: #ffffff;
  font-weight: bold;
}

#cart-table tbody tr:nth-child(even) {
  background-color: #f1f5f9;
}

/* Totals Box */
.bill-summary-card {
  background-color: #f8fafc;
  border: 1px solid #cbd5e1;
  border-radius: 6px;
  padding: 15px;
  margin-bottom: 20px;
  display: flex;
  flex-direction: column;
  gap: 6px;
  font-size: 15px;
}

.total-row {
  font-weight: bold;
  color: #333;
}

.discount-row {
  font-weight: bold;
  color: #ef4444; /* Red color for discount */
}

.final-row {
  font-size: 17px;
  font-weight: bold;
  color: #16a34a; /* Green color for final amount */
  margin-top: 4px;
}

/* Item Lists Container */
.lists-container {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 20px;
  margin-bottom: 20px;
}

.summary-section {
  background-color: #ffffff;
  border: 1px solid #cbd5e1;
  border-radius: 6px;
}

```

```

padding: 15px;
}

.summary-section h2 {
  font-size: 16px;
  color: #00008b;
  margin-top: 0;
  margin-bottom: 10px;
  border-bottom: 1px solid #cbd5e1;
  padding-bottom: 6px;
}

.summary-section ul {
  list-style-type: disc;
  padding-left: 20px;
  margin: 0;
  color: #4b5563;
  font-size: 14px;
}

.summary-section li {
  margin-bottom: 5px;
}

/* Footer Signature */
.footer {
  text-align: center;
  color: #64748b;
  font-size: 12px;
  margin-top: 15px;
  border-top: 1px solid #e2e8f0;
  padding-top: 12px;
}

/* Responsive Media Queries */
@media (max-width: 650px) {
  .controls-card {
    flex-direction: column;
    align-items: stretch;
  }
  .button-group {
    justify-content: space-between;
  }
  .button-group button {
    flex: 1;
    text-align: center;
  }
  .stats-grid {
    grid-template-columns: 1fr;
    gap: 10px;
  }
  .input-form {
    flex-direction: column;
  }
  .lists-container {
    grid-template-columns: 1fr;
  }
}

```

script.js

```

// =====
// 1. Tab switching navigation logic
// =====
document.addEventListener('DOMContentLoaded', () => {
  const tabFinder = document.getElementById('tab-finder');
  const tabCart = document.getElementById('tab-cart');
  const finderContainer = document.getElementById('finder-container');
  const cartContainer = document.getElementById('cart-container');

```



```

tabFinder.addEventListener('click', () => {
  tabFinder.classList.add('active');
  tabCart.classList.remove('active');
  finderContainer.classList.add('active');
  cartContainer.classList.remove('active');
});

tabCart.addEventListener('click', () => {
  tabCart.classList.add('active');
  tabFinder.classList.remove('active');
  cartContainer.classList.add('active');
  finderContainer.classList.remove('active');
});
});

// =====
// 2. Array Min & Max Finder Logic
// =====
let numbers = [];

// DOM Elements
const numberInput = document.getElementById('number-input');
const addNumberBtn = document.getElementById('add-number-btn');
const addRandomBtn = document.getElementById('add-random-btn');
const resetBtn = document.getElementById('reset-btn');
const pillsContainer = document.getElementById('array-pills-container');

const maxValDisplay = document.getElementById('max-val-display');
const minValDisplay = document.getElementById('min-val-display');
const totalElementsDisplay = document.getElementById('total-elements-display');

// Event Listeners
addNumberBtn.addEventListener('click', addNumber);
addRandomBtn.addEventListener('click', addRandom);
resetBtn.addEventListener('click', resetArray);

// Allow Enter key to trigger adding number
numberInput.addEventListener('keypress', (e) => {
  if (e.key === 'Enter') {
    addNumber();
  }
});

function addNumber() {
  const valStr = numberInput.value.trim();
  if (valStr === "") {
    alert("Please enter a valid number.");
    return;
  }

  const val = Number(valStr);
  if (isNaN(val)) {
    alert("Please enter a valid number.");
    return;
  }

  numbers.push(val);
  numberInput.value = '';
  numberInput.focus();

  updateFinderUI();
}

function addRandom() {
  // Generate random number between 1 and 100
  const randomNum = Math.floor(Math.random() * 100) + 1;
  numbers.push(randomNum);
}

```

```

    updateFinderUI();
}

function resetArray() {
    numbers = [];
    updateFinderUI();
}

function updateFinderUI() {
    // 1. Render array element badges/pills
    pillsContainer.innerHTML = '';
    numbers.forEach(num => {
        const pill = document.createElement('div');
        pill.className = 'pill';
        pill.textContent = num;
        pillsContainer.appendChild(pill);
    });

    // 2. Loop search for min/max
    if (numbers.length === 0) {
        maxValDisplay.textContent = '-';
        minValDisplay.textContent = '-';
        totalElementsDisplay.textContent = '0';
        return;
    }

    let min = numbers[0];
    let max = numbers[0];

    // Using standard for-loop to search minimum and maximum values
    for (let i = 1; i < numbers.length; i++) {
        if (numbers[i] < min) {
            min = numbers[i];
        }
        if (numbers[i] > max) {
            max = numbers[i];
        }
    }

    // Update display values
    maxValDisplay.textContent = max;
    minValDisplay.textContent = min;
    totalElementsDisplay.textContent = numbers.length;
}

// =====
// 3. Shopping Cart Calculator Logic
// =====
let cart = [];

// DOM Elements
const productNameInput = document.getElementById('product-name');
const productPriceInput = document.getElementById('product-price');
const productQtyInput = document.getElementById('product-qty');
const addProductBtn = document.getElementById('add-product-btn');
const cartTableBody = document.querySelector('#cart-table tbody');

const totalAmountEl = document.getElementById('total-amount');
const discountPercentEl = document.getElementById('discount-percent');
const discountAmountEl = document.getElementById('discount-amount');
const finalAmountEl = document.getElementById('final-amount');

const itemSummaryList = document.getElementById('item-summary-list');
const expensiveProductsList = document.getElementById('expensive-products-list');

// Add Product Event Listener
addProductBtn.addEventListener('click', addProduct);

```

```

// Allow pressing Enter key inside input fields to add a product
[productNameInput, productPriceInput, productQtyInput].forEach(input => {
  input.addEventListener('keypress', (e) => {
    if (e.key === 'Enter') {
      addProduct();
    }
  });
});

function addProduct() {
  const name = productNameInput.value.trim();
  const price = parseFloat(productPriceInput.value);
  const qty = parseInt(productQtyInput.value);

  // Validation checking
  if (!name) {
    alert("Please enter a valid product name.");
    return;
  }
  if (isNaN(price) || price <= 0) {
    alert("Please enter a valid price (greater than 0).");
    return;
  }
  if (isNaN(qty) || qty <= 0) {
    alert("Please enter a valid quantity (1 or more).");
    return;
  }

  // Create new product item object
  const item = {
    id: cart.length + 1,
    name: name,
    price: price,
    qty: qty,
    total: price * qty
  };

  // Push object to state array (Array Method: push)
  cart.push(item);

  // Reset Input fields
  productNameInput.value = '';
  productPriceInput.value = '';
  productQtyInput.value = '';
  productNameInput.focus();

  // Update UI Elements
  updateCartUI();
}

function updateCartUI() {
  // 1. Render Table Rows (Array Method: forEach)
  cartTableBody.innerHTML = '';
  cart.forEach(item => {
    const row = document.createElement('tr');
    row.innerHTML = `
      <td>${item.id}</td>
      <td>${item.name}</td>
      <td>${item.price.toFixed(2)}</td>
      <td>${item.qty}</td>
      <td>${item.total.toFixed(2)}</td>
    `;
    cartTableBody.appendChild(row);
  });

  // 2. Calculations (Array Method: reduce)
  const totalAmount = cart.reduce((accumulator, item) => {
    return accumulator + item.total;
  }, 0);
}

```

```

// Discount percentage rates
let discountRate = 0;
if (totalAmount >= 5000 && totalAmount < 20000) {
    discountRate = 0.05; // 5% discount
} else if (totalAmount >= 20000) {
    discountRate = 0.10; // 10% discount
}

const discountAmount = totalAmount * discountRate;
const finalAmount = totalAmount - discountAmount;

// Format currencies
totalAmountEl.textContent = `₹${totalAmount.toFixed(2)}`;
discountPercentEl.textContent = `${(discountRate * 100).toFixed(0)}%`;
discountAmountEl.textContent = `₹${discountAmount.toFixed(2)}`;
finalAmountEl.textContent = `₹${finalAmount.toFixed(2)}`;

// 3. Render Item Summary (Array Method: forEach)
itemSummaryList.innerHTML = '';
cart.forEach(item => {
    const li = document.createElement('li');
    li.textContent = `${item.name} : ${item.total.toFixed(2)}`;
    itemSummaryList.appendChild(li);
});

// 4. Filter and Render Expensive Products where Unit Price > 1000 (Array Method: filter &
forEach)
expensiveProductsList.innerHTML = '';
const expensiveItems = cart.filter(item => item.price > 1000);
expensiveItems.forEach(item => {
    const li = document.createElement('li');
    li.textContent = `${item.name} (${item.price.toFixed(2)})`;
    expensiveProductsList.appendChild(li);
});
}

```

4. Output:

PRN: 24070521038 | Name: Vansh Mahant | File Path: /practical5/index.html — Array Min/Max Finder (initial state)

🏠 Array Min/Max Finder

🛒 Shopping Cart Calculator

Array Min & Max Finder

⬆⬇⬆

Add NumberAdd RandomReset

Current Array

MAXIMUM VALUE

-

MINIMUM VALUE

-

TOTAL ELEMENTS

0

PRN: 24070521038 | Name: Vansh Mahant | File Path: /practical5/index.html — Array Min/Max Finder after adding numbers (42, 7, 99, 15)

🏠 Array Min/Max Finder

🛒 Shopping Cart Calculator

Array Min & Max Finder

⬆⬇⬆

Add NumberAdd RandomReset

Current Array

42

7

99

15

MAXIMUM VALUE

99

MINIMUM VALUE

7

TOTAL ELEMENTS

4

PRN: 24070521038 | Name: Vansh Mahant | File Path: /practical5/index.html — Shopping Cart Calculator with sample products

🔍 Array Min/Max Finder

🛒 Shopping Cart Calculator

Shopping Calculator

ID	Product	Price	Quantity	Total
1	Laptop	55000.00	1	55000.00
2	Mouse	500.00	2	1000.00
3	Keyboard	1500.00	1	1500.00

Total Amount : ₹57500.00

Discount (10%) : -₹5750.00

Final Amount : ₹51750.00

Item Summary

- Laptop : 55000.00
- Mouse : 1000.00
- Keyboard : 1500.00

Expensive Products (Price > 1000)

- Laptop (55000.00)
- Keyboard (1500.00)

5. Result/Conclusion:

A tabbed web page combining an Array Min/Max Finder and a Shopping Cart Calculator was successfully implemented using HTML, CSS and JavaScript. The finder tab dynamically builds an array from user input and computes the minimum and maximum using a standard for-loop. The cart tab uses array methods — push to add items, forEach to render rows and lists, reduce to compute the total amount, and filter to identify products priced above ₹1000 — and applies a tiered discount (5% for totals ≥ ₹5000, 10% for totals ≥ ₹20000). This experiment reinforced the concepts of array iteration, array methods (push, forEach, reduce, filter), and DOM manipulation in JavaScript.

Upload the Soft copy as well as complete code on GitHub link is mandatory.

File name: Practical 5_24070521038_Vansh Mahant_A_A1